BAS--102 APPLIED MATHEMATICS-I

L	T	P	Credits
3	1	0	3.5

Course Objective:

The course is aimed at developing the basic mathematical skills among the students of engineering that are imperative for effective understanding of engineering subjects. The main objective is to inculcate the knowledge of basic concepts of Calculus, Algebra, Complex Analysis and their applications for the solutions of engineering and mathematical problems. At the end of the course, students shall be able to deal with functions of several variables, matrices, system of linear equations, improper integrals and functions of complex variables. They shall learn how to implement their mathematical knowledge to solve real world problems.

Course Content

Section-A

Matrix Algebra: Special Matrices: Hermitian matrices, Skew-Hermitian matrices, Unitary matrices and Orthogonal matrices; Rank of matrices, Rank by Echelon form; Consistency of system of homogeneous and non-homogeneous equations: Gauss Elimination method, Gauss Jordan method; Eigen values and Eigen vectors of a matrix, Elementary properties of Eigen values and Eigen vectors,Eigen values of Special Matrices, Cayley Hamilton's Theorem.

Functions of several variables: Partial derivatives, Total differential, Approximation by Total Differentials, Derivatives of composite and implicit functions,Homogenous functions, Euler's theorem, Maximumand Minimum values of functions of two and three variables, Lagrange's Method of Multipliers.

Section-B

Complex Analysis: De Moivre's Theorem, Power and roots of complex numbers,Expansion of $\cos^n\theta$, $\sin^n\theta$ in terms of cosn θ and sinn θ and vice-versa, Analytic functions, Necessary and Sufficient condition for a function to be analytic, CR-equations, Polar form of CR-equations, Harmonic functions, Conjugate of Harmonic functions, Laplace equation.

Improper integrals: Improper integral of First and Second kind, Absolute convergence of Improper integrals, Comparison tests (without proof), Beta and Gamma functions and their properties.

RECOMENDED BOOKS:

1. R. K. Jain and S. R. K. Iyenger, Advanced Engineering Mathematics, Narosa Publishing House, New Delhi.
2. E. Kreyzic, Advanced Engineering Mathematics,Johan Wiley & Sons, (10^{th}Edition).
3. H. K. Dass, Advanced Engineering Mathematics, S. Chand and Company.
4. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher.

Scheme of Examination

- English will be the medium of instruction and examination.
- This course will carry 100 marks of which 50 marks shall be reserved for Internal Assessment and remaining 50 marks for external end semester examination.
- The duration of final written examination of this paper shall be of three hours.
- The students shall be declared passed in the paper if he/she secures minimum 40% marks in each of the Internal Assessment and External Examinations separately.

Instructions to the External Paper Setter

- The External Paper will carry 50 marks and would be of three hours. The Question paper will be divided into three Sections, namely Section-A, Section-B and Section-C. There will be FIVE questions in each of the Sections A and B, of five (05) marks each. However, a question may be segregated into subparts. Section C is compulsory and contains TEN (10) sub-parts each of two (2) marks. Candidates will be required to attempt SIX questions by selecting three Questions from each Sections A& B.
- The paper setter shall provide detailed marking instructions and solutions to numerical problems for evaluation purpose in the separate white envelopes provided for solutions.
- The paper setters should seal the internal & external envelope properly with signatures & cello tape at proper place.

ECE--102 BASIC ELECTRONICS ENGINEERING

L	T	P	Credits
3	1	0	3.5

Course Objective:

To enhance comprehension capabilities of student through understanding of electronic devices such as Diode, Transistor and FET. The student should also be able to acquire basic knowledge of Digital electronics and communication system.

Section-A

PN junction, Depletion layer, Barrier potential, Forward and reverse bias, Breakdown voltage, PIV, Characteristics of p-n junction diode, knee voltage, load line; and operating Point. Ideal p-n junction diode, junction capacitance, zener diode. Rectifiers and filters-Half wave, centre tap full wave and bridge rectifier, clipping and clamping circuit, voltage regulation.

BJT - Introduction, Basic theory of Operation of PNP ad NPN transistor, V-I characteristics, CB, CE and CC configuration, Basic BJT Amplifiers. Introductory idea of RC multistage amplifiers. FET- Introduction, V-I characteristics and operation, UJT - Introduction, V-I characteristics and operation.

Section-B

Number Systems: Number systems, Conversions, Number Representations, Demorgan 's Theorem, Boolean Algebra and Arithmetic operations. Binary codes-Grey codes, Excess-3 code, BCD code, symbols and truth tables of NOT Gate, AND Gate, NAND Gate, OR Gate, NOR Gate, EX-OR Gate and EX-NOR Gate.

Introduction to communication system, General block diagram, need for communication, need of modulation and demodulation, Block diagram of radio transmission and reception system and function of each block.

Course Learning Outcomes (CLO): After the completion of the course, the student should be able to:

1. Demonstrate the use of semiconductor diode in various applications.
2. Discuss and explain the working of transistor ,their configuration and application.
3. Recognize and apply the number system and Boolean algebra.
4. Define the communication system and differentiate various modulation techniques.
5. Explain radio transmission and reception.

Text Books:

1. Milliman, J. and Halkias, C.C., Electronic Devices and Circuits, Tata McGraw Hill, (2007).
2. Boylestad, R.L. and Nashelsky, L., Electronic Devices & Circuit Theory, Perason (2009).
3. Digital Design by Morris Mano, PHI, 4th edition, 2008.
4. Digital principles and Applications, by Malvino Leach, TMH, 2011.
5. Electronic Communication Systems by G. Kennedy And B. Davis, Mc Graw Hill, 4th Edition, (2006).

Reference Books:

1. Edward Hughe, Electrical Technology, Addison-Wisley, New York.
2. Naidu and S. Kamakshaiah, Introduction to Electrical Engineering Tata McGraw Hills, New Delhi.
3. V. Deltoro, Principle of Electrical Engineering, Tata McGraw Hill, New Delhi.
4. Smith and Dorf, Circuits Devices and Systems, John Wiley and Sons.
5. S Thomas L. Gloyd, Electronics Fundamentals Circuits, Devices Applications, Prentice Hall International Inc.
6. B.P Lathi, Communication systems Engg., Pearsons.

MCE--102 MANUFACTURING PROCESSES

L	T	P	Credits
3	0	0	3.0

Course Objective:

The objective of this course is to have a good understanding the basic concepts of manufacturing via engineering materials, casting, machining, forming, joining, welding and assembly, enabling the students to develop a basic knowledge of the mechanics, operation and limitations of basic machining tools. The course also introduces the concept of basic carpentry operations.

At the end of the course, the students should be able to - (1) demonstrate the capability of selecting suitable manufacturing processes to manufacture the products optimally, (2) ability to clear basic fundamental concepts of machining, welding, casting, forming processes (3) selecting or suggesting suitable manufacturing processes to achieve the required products with the aim of avoiding material and time wastage.

Section – A

Introduction: Common engineering materials and their important mechanical and manufacturing properties. General classification of manufacturing processes, Importance of manufacturing processes, economics and selection of manufacturing processes.

Metal Casting: Principles of metal casting, casting terminology, Patterns, their functions, types, materials and pattern allowances, Characteristics of molding sand, Types of sand molds, Types of cores, chaplets and chills; their materials and functions. Casting Defects, their causes and remedies.

Machining Processes: Principles of metal cutting, cutting tool materials and applications, types of single point cutting tools. Geometry of single point cutting tool. Cutting fluids and their functions, types of cutting fluids, selection of cutting fluids.

Machine Tools: Introduction to Centre Lathe, parts of a lathe, lathe attachments, Operations performed on lathe, work holding devices in Lathes.

Section – B

Introduction to shaper and planner machines, their operations. Introduction to multipoint cutting tools. Introduction to milling and milling operations, Drilling and allied operations, Sawing operations.

Welding & Allied Joining Processes: Welding classification, types of welding electrodes,functions of flux and types of welding joints. Elements of Electric arc, Gas, Resistance and Thermit welding.Soldering, Brazing and Braze welding, Submerged arc welding (SAW). Applications of various welding processes. Welding defects, their causes and remedies.

Metal Forming and Shearing: Hot and cold working, Types of Forging processes. Rolling, Wire drawing and extrusion processes, drawing, bending, spinning, stretching, embossing and coining. Die and punch operation, shearing, piercing and blanking, notching, lancing, bending and deep drawing operations.

Carpentry Operations: Woods and their types, seasoning of wood, types of joints.

Recommended Books

1. Degramo, Kohser and Black. Materials and Processes in Manufacturing, 8thEdition, Prentice Hall of India, New Delhi.
2. Amstead Ostwald, and Bageman, Manufacturing Processes, John Wiley and sons, New Delhi.
3. Campbell, Principles of Manufacturing, Materials and Processes, Tata Macgraw HillCompany
4. Kalpakjian, S. and Schemid, S.R., Manufacturing Engineering & Technology,Prentice Hall, New York.
5. Groover, M.P., Fundamentals of Modern manufacturing: Materials, Processes andSystems, John Wiley and Sons Inc., New York.
6. B. S. Raghuwanshi, Workshop Technology (Part – I & II), Dhanpat Rai and Co., New Delhi.
7. Singh, Manufacturing Technology, Pearson Education Asia, New Delhi.
8. Khanna, O.P. and Lal, M., A Text Book of Production Technology, Dhanpat Rai Publication, New Delhi.

BAS --101 APPLIED PHYSICS – I

L	T	P	Credits
3	1	0	3.5

Course Objective:
The course is aimed at developing the basic scientific skills among students of engineering that are imperative for effective understanding of engineering subjects. Physical concepts included in this syllabus are importance for understanding various engineering and technological problems. At the end of the course, students shall be able to deal with phenomena related to oscillations, diffraction, interference, polarization, lasers and quantum mechanics. They will learn how to implement their scientific knowledge to solve real world problems.

Section A

Simple harmonic motion: Differential equation, Energy of simple harmonic oscillator, lissajous figures formed by superposition of two SHM, Charge oscillations in a LC circuit. Damped oscillator: Differential equation, methods of describing damping of an oscillator- logarithmic decrement, relaxation time, quality factor, Damped oscillations in a LCR circuit.
Interference by division of amplitude: plane parallel thin films, colors in thin films, non-reflecting films/coatings, high reflectivity thin film coatings, Michelson interferometer.
Fraunhoffer diffraction from circular aperture, double slit and a grating (normal incidence case), Rayleigh's criteria of resolution, resolving power of telescope, microscope and grating.
Polarization by double refraction, Nicol prism, Concept of plane, circular and elliptical polarization with mathematical expression.

Section B

Transitions between energy states, Einstein coefficients, principle and properties of laser beam, three and four level lasers, elementary description of principle, construction and operation of He-Ne laser, CO_2 laser, ruby laser, Nd-YAG laser and semi-conductor laser. Applications of lasers.
Propagation of light through optical fiber, its geometry, numerical aperture and acceptance angle, step index and graded index fibers, Signal attenuation and dispersion (qualitative ideas). Applications of optical fibers.
de-Broglie waves, velocity of de-Broglie waves, wave packets, Davisson and Germer experiment (Qualitative only), wave functions, time dependent and time independent Schrodinger wave equation, expectation value, application of Schrodinger equation to particle in an infinite potential box, potential barrier (tunneling effect)

Recommended Books

1. Wave and Vibrations by H.J. Pains
2. Fundamentals of optics by Jenkins and White (McGraw Hills)
3. Lasers- Theory and applications by Thyagrajan and Ghatak (McMillan Publishers)
4. Physics for Engineering Applications by S. Puri (Narosa Publishers)

Scheme of Examination

- English will be the medium of instruction and examination.
- This course will carry 100 marks of which 50 marks shall be reserved for Internal Assessment and remaining 50 marks for external end semester examination.
- The duration of final written examination of this paper shall be of three hours.
- The students shall be declared passed in the paper if he/she secures minimum 40% marks in each of the Internal Assessment and External Examinations separately.

Instructions to the External Paper Setter

- The External Paper will carry 50 marks and would be of three hours. The Question paper will be divided into three Sections, namely Section-A, Section-B and Section-C. There will be FIVE questions in each of the Sections A and B, of five (05) marks each. However, a question may be segregated into subparts. Section C is compulsory and contains TEN (10) sub-parts each of two (2) marks. Candidates will be required to attempt SIX questions by selecting three Questions from each Sections A& B.
- The maximum limit on numerical problems to be set in the paper is 35%.
- The paper setter shall provide detailed marking instructions and solutions to numerical problems for evaluation purpose in the separate white envelopes provided for solutions.
- The paper setters should seal the internal & external envelope properly with signatures & cello tape at proper place.
- Use of non-programmable calculator shall be specified clearly if required.

GROUP A: SECOND SEMESTER & GROUP B: FIRST SEMESTER

BUILDING MATERIALS

Course Code: CVLB1106T / CVLB1206T

L	T	P	Credits
3	0	0	3.0

Course Objectives: The course aims to: 1. Provide understanding of properties, classification, and manufacturing of basic construction materials such as bricks, aggregates, lime, and mortar; 2. Develop knowledge of concrete technology, including production, properties, and quality control; 3. Introduce manufacturing and properties of cement and its role in construction; 4. Familiarize students with properties, defects, and uses of timber and stones; 5. Provide understanding of metals and miscellaneous construction materials used in engineering applications.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Identify and evaluate properties of common construction materials; 2. Understand production and quality control of concrete and cement; 3. Analyze characteristics and applications of timber, stones, and metals; 4. Select appropriate materials for specific construction applications; 5. Apply knowledge of various building materials in engineering practice.

Section A

Bricks: Composition of good brick earth, harmful ingredients, manufacture of bricks, characteristics of good brick, shapes, classification of bricks as per IS 1077-1985 and testing.
Aggregates: Classification, Characteristics, soundness of aggregates, fineness modulus.
Lime and Mortar: Classifications & properties.
Concrete: Introduction, Properties of concrete, water cement ratio, workability, compressive strength, grades, production of concrete: Batching, mixing, transportation, placing, compaction and curing of concrete, quality control of concrete, concrete mix design.

Section B

Cement: Manufacture, basic properties of cement compounds, grades, packing storage, quality control and curing.
Timber: Classification and identification of timber, defects in timber, characteristics of good timber, uses and testing of timber, seasoning of timber.
Stones: Classification of rocks, test for stones, characteristics of a good building stone, deterioration of stones, common building stones of India.
Metals: Manufacture of steel, market forms of steel e.g. mild steel and HYSD steel bars, rolled steel sections, Thermo Mechanically Treated (TMT) bars.
Miscellaneous Materials: Insulating materials, Paints and varnishes; materials for doors and windows, Asphalt and Bitumen

References Books:

1. Rangawala, S. C., Engineering Materials, Charotar Publishing House (1992).
2. Gambhir, M. L., Concrete Technology, Tata McGraw Hill Publishing Co. Ltd. (2004).
3. Kumar, Sushil, Engineering Materials, Metropolitan Press (1994).
4. Kumar, Sushil, Building Construction, Standard Publishers and Distributors (1990).
5. Punmia B.C., Building Construction, Laxmi Publishing House (1993).

BAS--151 APPLIED PHYSICS – I LAB

L	T	P	Credits
0	0	2	1.0

Course Objective:

The main aim of the Applied Physics –I Lab is to inculcate the practical abilities in the students along with theoretical studies. Students will be able to understand the concept of electromagnetic waves, phenomena of reflection, refraction, interference and diffraction of electromagnetic light through various experiments. At the end, students will be able to understand the optical properties of waves.

List of Experiments

1. To measure the wavelength of Laser (He-Ne) light by using reflection grating.
2. To measure angle of prism using a spectrometer.
3. To measure refractive index of prism using a spectrometer.
4. To determine wavelength of sodium light using a plane diffraction grating.
5. To determine specific rotation of sugar using Polarimeter.
6. To study Transverse nature of light.
7. To study use of CRO to measure amplitude and frequency of different waveforms.
8. To superposition of two waves using Lissajous figures.
9. To determine numerical aperture of an optical fibre.
10. To study optical fibre transmitter & receiver function for audio signal.

Scheme of Examination

- English will be the medium of instruction and examination.
- This course will carry 100 marks of which 50 marks shall be reserved for Internal Assessment and remaining 50 marks for external end semester examination.
- The duration of final written examination of this paper shall be of two hours.
- The students shall be declared passed in the paper if he/she secures minimum 40% marks in the Internal Assessment and end semester external examinations collectively.

Instructions to the Paper Setter

- The External Paper will carry 50 marks and would be of two hours.
- Use of non-programmable calculator shall be specified clearly if required.

MCE--152 MANUFACTURING PROCESSES LAB

L	T	P	Credits
0	0	3	1.5

Course Objective:

The objective of this course is to inculcate practical knowledge to students regarding basic manufacturing processes like: casting, machining, sheet metal forming, fitting, smithy, welding and basic carpentry operations.

At the end of the course, the students should be able to - (1) demonstrate the capability of selecting suitable manufacturing processes to manufacture the products optimally, (2) ability to clear basic fundamental concepts of machining, welding, casting, smithy, sheet metal forming, fitting and carpentry processes.

List of Experiments

1. Machine Shop: Six in one job in Machine Shop (involving turning, step cutting, threading, grooving, taper turning, knurling, drilling and tapping)
2. Fitting Shop: L – Cutting from square piece in fitting shop (involving squaring, L – cutting and squaring, drilling, tapping, reaming)
3. Sheet Metal Shop: Layout marking, cutting/shearing, bending in box shape with drilling and Riveting
4. Carpentry Shop: Cross and Lap joints, T – Joint
5. Welding Shop: Butt Welding / Gas welding, Soldering.
6. Foundry Shop: Moulding of Flange, Moulding of Core and casting of pipe.
7. Smithy Shop: Poker, Circular Ring.

ECE--153 ELECTRICAL AND ELECTRONICS LAB

L	T	P	Credits
0	0	2	1.0

Objective: The educational objective of the **Electrical and Electronics Lab** is to provide student the basic engineering knowledge by way of electrical and electronic devices and components.

List of Experiments

1. Identification and familiarization with the basic tools used in lab.
2. Familiarization and testing of Resistance, Capacitor & Inductors.
3. To study various types of switches such as normal/miniature toggle, slide, push button, rotary, micro switches, SPST, SPDT, DPST, DPDT, band selector, multiway Master Mains Switch.
4. To study various types of protective devices such as Wire fuse, cartridge fuse, slow acting/fast acting fuse, HRC fuse, and thermal fuse, single/multiple circuit breakers, over and under current relays.
5. To get familiar with the working knowledge of the measuring instruments: a) Ammeter & Voltmeter b) Cathode ray oscilloscope (CRO) c) Multimeter (Analog and Digital)
6. To get familiar with the working knowledge of the following instruments: a) Signal generator b) Function generator c) Power supply.
7. Familiarization and testing of Diode, BJT & FET.
8. Use of diode as half wave and full wave rectifier.
9. To verify Kirchhoff's laws.
10. Verification of truth tables of logic gates.
11. Fabrication of Printed Circuit Board.
12. To learn soldering and desoldering techniques.

GROUP A: SECOND SEMESTER & GROUP B: FIRST SEMESTER

BUILDING MATERIALS LAB

Course Code: CVLB1156P / CVLB1256P

L	T	P	CREDITS
0	0	2	1.0

Course Objectives: The course aims to: 1. Provide understanding of building planning, construction practices, and bye-laws; 2. Introduce techniques of damp proofing and fire resistance; 3. Familiarize students with properties and testing of basic construction materials.

Course Outcomes (COs): After completion of this course, the students will be able to: 1. Identify factors affecting planning and construction of buildings; 2. Apply construction practices and comply with building bye-laws; 3. Explain and implement damp proofing and fire resistance techniques; 4. Perform standard tests on cement, fine aggregates, coarse aggregates, and bricks; 5. Analyze test results and relate material properties to construction applications.

List of Experiments:

- Perform tests on Cement
- Perform tests on Fine Aggregates
- Perform tests on Coarse Aggregate
- Perform tests on Bricks.